

UniNotes

COMPUTER SCIENCE

Database Systems Revision Notes

Relational databases, keys, normalisation, SQL and transaction fundamentals.

How to use these notes

Start with the key definitions, then use the tables/diagrams to connect ideas. Finish with the exam checklist. The aim is high information density: short wording, clear concepts, no filler.

1. Relational model

A relational database stores data in tables (relations). Rows are records/tuples; columns are attributes. A schema defines structure, data types and constraints.

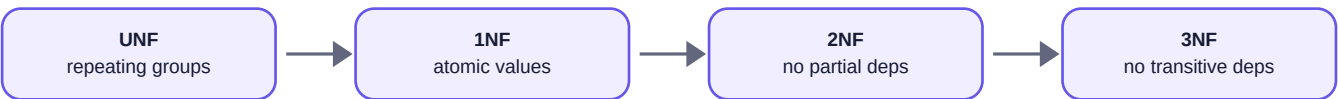
Primary key	Uniquely identifies each row; should be stable and non-null.
Foreign key	References a key in another table to create a relationship.
Candidate key	Minimal attribute set that could uniquely identify a row.
Composite key	A key made from multiple columns.

2. Relationships

One-to-one: one row maps to at most one row. One-to-many: the foreign key normally sits on the "many" side.

Many-to-many: use a junction table containing foreign keys to both tables.

3. Normalisation



1NF removes repeating groups and requires atomic values. 2NF removes partial dependency on part of a composite key. 3NF removes transitive dependency where a non-key attribute depends on another non-key attribute.

Why normalise?

Main benefits: less duplicated data, fewer update/insert/delete anomalies, and clearer integrity rules. Over-normalisation can increase joins, so design still depends on workload.

4. SQL essentials

Read	SELECT columns FROM table WHERE condition;
Create row	INSERT INTO table (...) VALUES (...);
Update	UPDATE table SET col=value WHERE ...;
Delete	DELETE FROM table WHERE ...;
Join	SELECT ... FROM A JOIN B ON A.key=B.fk;

5. Transactions and ACID

Atomicity	All steps happen or none do.
Consistency	Rules/constraints remain valid.
Isolation	Concurrent transactions do not interfere incorrectly.
Durability	Committed changes survive failures.

Exam-ready checklist

- Identify primary and foreign keys.
- Model one-to-many and many-to-many relationships.
- Explain 1NF, 2NF and 3NF.
- Write basic SELECT/INSERT/UPDATE/DELETE statements.
- Explain each ACID property in one sentence.

Fast recall

If you cannot explain a concept in one or two sentences without looking, revisit that section. Aim to recall the definition, mechanism and one example.

Uni Notes keeps revision focused: concise explanations, useful detail, and diagrams only where they improve understanding.